SPANS-08: Cost-Efficient DQL Queries

> **Series:** SPANS — Distributed Tracing and Spans | **Notebook:** 8 of 8 | **Created:** December 2025 | **Last Updated:** 10/28/2026

Optimizing Span Queries for Performance and Cost Efficiency

This notebook covers best practices for writing cost-efficient DQL queries, minimizing query consumption while maintaining effective observability.

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10. Series Complete! 🎉

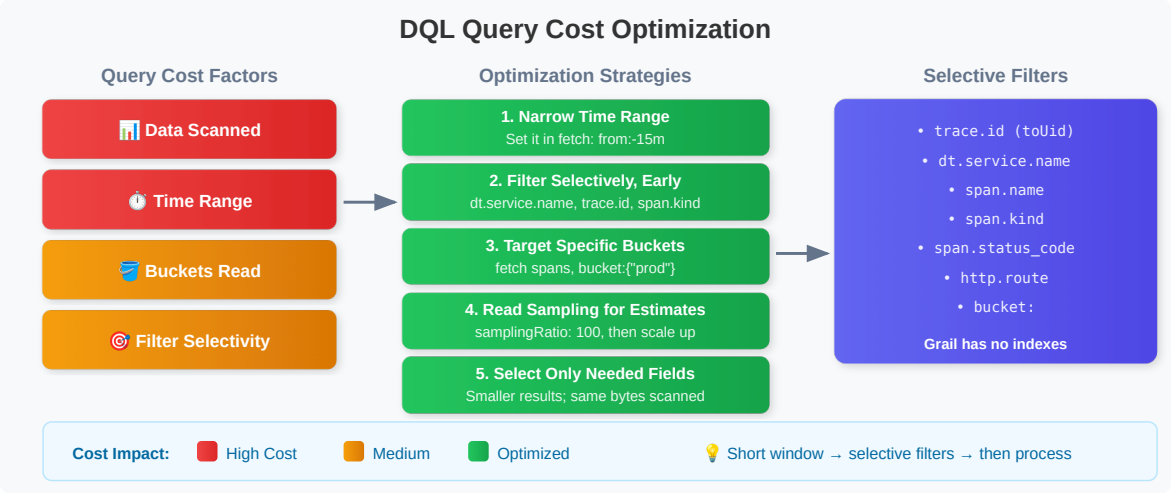
Prerequisites

Before starting this notebook, ensure you have:

- ✅ Completed previous SPANS notebooks (01-07)
- ✅ Understanding of DQL fundamentals
- ✅ Access to span data in your tenant (`storage:spans:read`)

1. Understanding Query Costs

Query cost is driven by how much data a query scans. Under the **Dynatrace Platform Subscription (DPS)**, span queries bill under the **Traces - Query** capability, metered in bytes scanned. Every query result reports its `scannedBytes`, so you can measure each change below instead of guessing.



Cost Optimization Priorities

Priority	Strategy	Impact
1	Limit time range	High — 1 h → 15 min cut scanned bytes from 910 MB to 234 MB
2	Filter selectively, and early	High — `dt.service.name == "checkout"` scanned 54 MB of the same hour
3	Query specific buckets	Medium — targets data
4	Read sampling for estimates	Medium — `samplingRatio` reads a fraction and you scale up
5	Select only needed fields	Low — smaller results; scanned bytes unchanged

Priority	Strategy	Impact
6	Use aggregations	Low — smaller results

All figures: one hour of spans on the validation tenant, 10/02/2026.

> **Sources:** DQL best practices (DT docs) — *"A shorter analysis window provides better performance based on identical data sets."*

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2. Filter Early Pattern

The **most important** optimization: Apply filters as early as possible in the query pipeline.

```
// ✅ EFFICIENT: Filter BEFORE any processing
fetch spans, from:-1h
| filter span.kind == "server"
| filter dt.service.name == "checkout"   // replace with one of your services
| filter span.status_code == "error"
| fields start_time, span.name, duration
| limit 100
```

```
// ⚠️ LATE FILTER: processing and aggregation before the filter.
// On 10/02/2026 Grail's optimizer pushed this filter down and both forms
scanned
// ~54 MB — but that is the optimizer's choice, not a guarantee. Filter first
so the
// cost does not depend on it, and so the intent is obvious to the next
reader.
fetch spans, from:-1h
| fieldsAdd duration_ms = duration / 1ms
| summarize {avg_duration = avg(duration_ms)}, by:{dt.service.name, span.name}
| filter dt.service.name == "checkout"
| limit 100
```

```
// ✅ EFFICIENT: Combine multiple filter conditions
fetch spans, from:-1h
| filter span.kind == "server"
        and isNotNull(dt.service.name)
        and duration &gt; 100ms
| fields start_time, span.name, duration
| sort duration desc
| limit 50
```

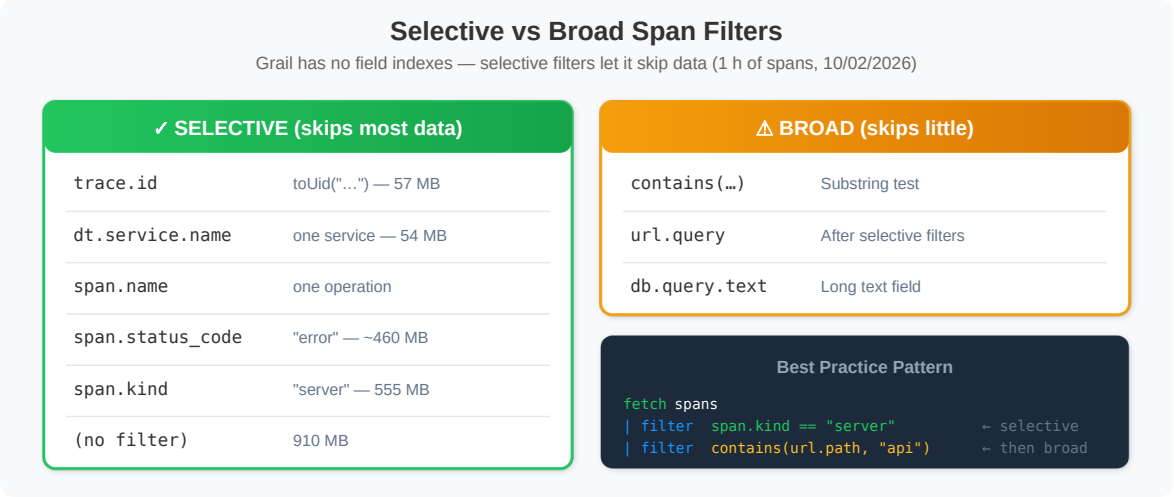
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3. Selective Filters and Data Skipping

Grail has **no field indexes**. Its storage technology *"obsoletes indexes"*, and query processing works by parallel scanning, *"eliminating the need for indexes"*. What makes a filter cheap is **selectivity**: when a condition rules out most of the data, Grail can skip reading it, and `scannedBytes` falls.

Measured on one hour of spans (10/02/2026):

Filter	Bytes scanned
none (`summarize count()`)	910 MB
`span.kind == "server"`	555 MB
`span.status_code == "error"`	~460 MB
`trace.id == toUid("…")`	57 MB
`dt.service.name == "checkout"`	54 MB



> 💡 **Tip:** `trace.id` is a `uid`, not a string. `trace.id == "abc…"` is **always false** — Grail says so in an INFO and a SEVERE notification, but returns zero rows rather than an error. Write `trace.id == toUid("abc…")`.
>
> **Sources:** Dynatrace Grail architecture (DT docs). **Dictionary:** `trace.id` (`stable`, `uid`), read 10/02/2026.

```
// ✅ FAST: Selective filters first
fetch spans, from:-1h
| filter isNotNull(dt.service.name)
| filter span.kind == "server"
| filter span.status_code == "error"
| fields start_time, span.name, duration
| limit 100
```

```
// ⚠️ BROAD: a substring test on its own skips little data
fetch spans, from:-1h
| filter contains(url.path, "checkout")  // broad
| limit 100
```

```
// ✅ BETTER: Selective filters first, then the substring test
fetch spans, from:-1h
| filter span.kind == "server"
| filter isNotNull(dt.service.name)
| filter contains(url.path, "checkout")  // substring test after the selective
filters
| limit 100
```

4. Field Selection

Select only the fields you need - avoid fetching all attributes.

```
// ✅ EFFICIENT: Select only required fields
fetch spans, from:-1h
| filter span.kind == "server"
| fields start_time,
         dt.service.name,
         span.name,
         duration,
         span.status_code
| limit 100
```

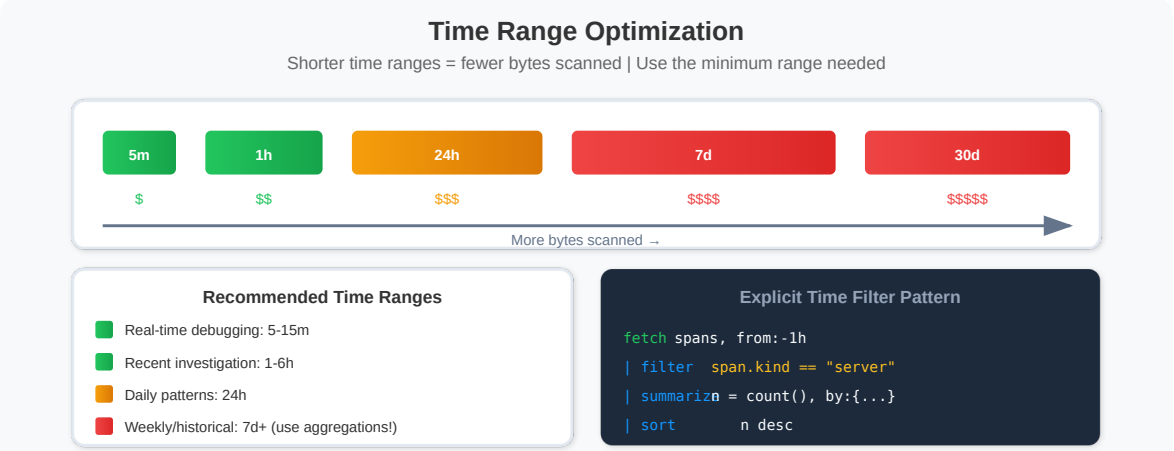
```
// ❌ INEFFICIENT: Fetching all fields (default behavior)
// Returns ALL fields for each span - more data transfer
fetch spans, from:-1h
| filter isNotNull(dt.service.name)
| limit 5
```

```
// ✅ EFFICIENT: Compute fields only when needed
fetch spans, from:-1h
| filter span.kind == "server"
| filter duration &gt; 500ms
| fieldsAdd duration_ms = duration / 1ms
| fields start_time, dt.service.name, span.name, duration_ms
| sort duration_ms desc
| limit 50
```

---

5. Time Range Optimization

Always use the smallest time range that meets your needs.




```
// Narrow window set in fetch. (A start_time filter right after fetch is
pushed down to the
// same window — measured 10/02/2026 — but from: says it directly.)
fetch spans, from:-15m
| filter span.kind == "server"
| summarize {span_count = count()}, by:{dt.service.name}
| sort span_count desc
| limit 20
```

```
// Narrow time range for troubleshooting specific incident
fetch spans, from:-30m
| filter span.kind == "server"
| filter span.status_code == "error"
| fields start_time, dt.service.name, span.name, span.status_message
| sort start_time desc
| limit 100
```

```
// Use aggregations for longer time ranges to reduce output
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    span_count = count(),
    error_count = countIf(span.status_code == "error" or
http.response.status_code &gt;= 500),
    avg_duration_ms = avg(duration) / 1ms
  }, by:{dt.service.name}
| sort span_count desc
| limit 25
```

6. Aggregation Efficiency

Use aggregations to summarize data instead of retrieving raw records. Simpler aggregations are faster.

```
// ✅ EFFICIENT: Basic aggregations
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    requests = count(),
    errors = countIf(span.status_code == "error" or http.response.status_code
&gt;= 500)
  }, by:{dt.service.name}
| sort requests desc
| limit 20
```

```
// ✅ EFFICIENT: Summarize at source with error rate calculation
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    total_requests = count(),
    error_count = countIf(span.status_code == "error" or
http.response.status_code &gt;= 500),
    p50_duration_ms = percentile(duration, 50) / 1ms,
    p95_duration_ms = percentile(duration, 95) / 1ms
  }, by:{dt.service.name}
| fieldsAdd error_rate_pct = (error_count * 100.0) / total_requests
| sort total_requests desc
| limit 20
```

```
// ⚠️ More percentiles add compute time — they do not change bytes scanned,
the billed
// quantity. Ask for the ones you will use.
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    p50 = percentile(duration, 50) / 1ms,
    p75 = percentile(duration, 75) / 1ms,
    p90 = percentile(duration, 90) / 1ms,
    p95 = percentile(duration, 95) / 1ms,
    p99 = percentile(duration, 99) / 1ms
  }, by:{dt.service.name}
| limit 20
```

```
// Time-bucketed aggregations for trends
fetch spans, from:-1h
| filter span.kind == "server"
| fieldsAdd time_bucket = bin(start_time, 5m)
| summarize {
    request_count = count(),
    avg_duration_ms = avg(duration) / 1ms
  }, by:{time_bucket, dt.service.name}
| sort time_bucket desc, request_count desc
| limit 100
```

---

7. High-Cardinality Grouping

Avoid grouping by high-cardinality fields (fields with many unique values).

```
// ❌ BAD: Grouping by high-cardinality field
// trace.id could have millions of unique values!
fetch spans, from:-1h
| summarize {trace_span_count = count()}, by:{trace.id}
| limit 10
```

```
// ✅ GOOD: Group by lower-cardinality fields
fetch spans, from:-1h
| summarize {span_count = count()}, by:{dt.service.name, span.name}
| sort span_count desc
| limit 20
```

```
// ✅ GOOD: If you need trace analysis, filter first
fetch spans, from:-15m                    // Narrow time
| filter span.status_code == "error"      // Reduce first
| summarize {
    span_count = count(),
    services = collectDistinct(dt.service.name)
  }, by:{trace.id}
| sort span_count desc
| limit 20
```

---

8. Production Query Patterns

Optimized query templates for common production use cases.

```
// Production Pattern: Service Health Dashboard
// Optimized for dashboard refresh (low cost)
fetch spans, from:-1h, bucket: {"default_spans"}   // use your own bucket
names
| filter span.kind == "server"
| summarize {
    requests = count(),
    errors = countIf(span.status_code == "error" or http.response.status_code
&gt;= 500),
    avg_latency_ms = avg(duration) / 1ms
  }, by:{dt.service.name}
| fieldsAdd error_rate = (errors * 100.0) / requests
| sort requests desc
| limit 20
```

```
// Production Pattern: Error Investigation
// Fast targeted query for debugging
fetch spans, from:-30m, bucket: {"default_spans"}   // use your own bucket
names
| filter span.kind == "server"
| filter span.status_code == "error"
| fields start_time,
         dt.service.name,
         span.name,
         span.status_message,
         trace.id
| sort start_time desc
| limit 50
```

```
// Production Pattern: Latency Trend Analysis
// Using bin() for time-series without makeTimeseries
fetch spans, from:-1h, bucket: {"default_spans"}   // use your own bucket
names
| filter span.kind == "server"
| filter isNotNull(dt.service.name)
| fieldsAdd time_bucket = bin(start_time, 5m)
| summarize {
    request_count = count(),
    p95_latency_ms = percentile(duration, 95) / 1ms
  }, by:{time_bucket}
| sort time_bucket desc
| limit 50
```

```
// Production Pattern: Top Slow Operations
// Focus on actionable data
fetch spans, from:-1h, bucket: {"default_spans"}   // use your own bucket
names
| filter span.kind == "server"
| filter duration &gt; 1s
| summarize {
    occurrence_count = count(),
    avg_duration_ms = avg(duration) / 1ms,
    max_duration_ms = max(duration) / 1ms
  }, by:{dt.service.name, span.name}
| sort avg_duration_ms desc
| limit 20
```

```
// Production Pattern: Well-optimized comprehensive query
// Follows all optimization rules
fetch spans, from:-1h                      // 1. Appropriate time range
| filter span.kind == "server"             // 2. Filter early
| filter span.status_code == "error"       // 3. Selective filter
| fields start_time, span.name, duration, http.route  // 4. Only needed fields
(smaller results)
| summarize {
    error_count = count(),
    avg_duration_ms = avg(duration) / 1ms
  }, by:{span.name, http.route}            // 5. Low cardinality
| sort error_count desc
| limit 20                                 // 6. Limited results
```

9. Performance Checklist

Use this checklist before running production queries:

```
Query Optimization Checklist:

□ 1. Time range - Is it as short as possible for my needs?
□ 2. Filter early - Are filters immediately after fetch?
□ 3. Selectivity - Do my first filters rule out most of the data?
      • trace.id (toUid!), dt.service.name, span.name
      • span.kind, span.status_code
□ 4. Field selection - Am I selecting only needed fields?
□ 5. Low cardinality - Are my group-by fields low cardinality?
□ 6. Result limits - Do I have appropriate limits?
□ 7. Bucket targeting - Am I querying specific buckets if known?
```

Summary

In this notebook, you learned:

✅ **Query cost factors** and optimization priorities
✅ **Filter early pattern** to reduce data scanned
✅ **Selective filters** — Grail has no indexes; selectivity drives data skipping
✅ **Field selection** to minimize data transfer
✅ **Time range optimization** for cost control
✅ **Aggregation efficiency** for summarized results
✅ **High-cardinality grouping** pitfalls to avoid
✅ **Production patterns** for common use cases
✅ **Performance checklist** for query review

Series Complete! 🎉

You have completed the **Spans & Distributed Tracing** notebook series!

What You've Learned:

1. **Fundamentals** - Span structure and distributed tracing concepts
2. **Querying** - DQL syntax for effective span analysis
3. **Troubleshooting** - Error detection and root cause analysis
4. **Topology** - Service dependencies and flow visualization
5. **Analytics** - Advanced metrics and trend analysis
6. **Security** - Security monitoring and compliance with spans
7. **Buckets & Pipeline** - Data architecture, OpenPipeline, and governance
8. **Cost Optimization** - Time range, selective filters, buckets and read sampling

Next Steps:

- Apply these patterns to your own Dynatrace environment
- Build dashboards using the optimized query patterns
- Configure alerts based on span analytics
- Set up OpenPipeline for data optimization
- Explore Dynatrace Intelligence integration for intelligent analysis

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References

- DQL best practices (DT docs)
- Dynatrace Grail architecture (DT docs)
- Data source commands — fetch (DT docs)

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.